

Compound Interest and Depreciation



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One formula does it all: $\text{amount} = \text{start} \times \text{multiplier}^{\text{years}}$. Growth has a multiplier above 1, depreciation below 1.

THE TOPIC ON ONE MAP

Compound Interest and Depreciation

01 Compound growth

savings for 3 years

value increases 4% a year

"total interest"

02 Depreciation

car loses 15% a year

total % lost in 2 years

band changed?

03 Compare and changing rates

which bank?

1.8% then $x\%$

compound vs simple

04 Work backwards: rate, start or time

find x from the total

find P

least number of years

THE FORMULA Read the story, fill three boxes:

$$\begin{array}{c}
 \boxed{\text{amount}} = \boxed{\text{start}} \times \boxed{\text{multiplier}}^{\text{years}} \\
 \text{up } r\%: 1 + \frac{r}{100} \quad \text{down } r\%: 1 - \frac{r}{100}
 \end{array}$$

01 Compound Growth

Interest is added to the new amount each year, so the multiplier is used again and again: multiply once, with a power.

WHEN YOU SEE → YOU DO

“compound interest”, “increases by 4% each year”

→ $\text{start} \times 1.04^n$

“how much money will she have”

→ the amount (start + interest)

“total amount of interest”

→ amount – start

“to the nearest dollar / 3 s.f.”

→ round only at the end

Multiplier

Power = years

Start $\times$ multiplierⁿ

Round

WORKED MODEL • NOVEMBER 2024 P2H

Nisha invests 20 000 euros for 3 years at 3.5% per year compound interest. How much will she have?

1 multiplier 1.035

up 3.5%

2 $20\,000 \times 1.035^3 = 22\,174.36$

power 3

ANSWER 22 174 euros

× **TRAP** Simple interest: $20\,000 \times 0.035 \times 3$ only adds interest on the first amount. “Compound” uses the power.

IDEA 01

Practise: Compound Growth

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Growth

CH 1 • TOPIC 7 3 marks

£5000 is invested for 2 years at 3% per year compound interest. Work out the value at the end.

SELECTED PRACTICE**Q2** **PRACTICE** Practice • Medium

CH 1 • TOPIC 7 2 marks

- 1 Toby invested £7500 for 2 years in a savings account. He was paid 4% per annum compound interest.

How much money did Toby have in his savings account at the end of 2 years?

(2 marks)

Q3**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 6 marks

5 (a) Derek buys a house for £150 000

He sells the house for £154 500

Work out Derek's percentage profit.

(3 marks)

(b) Derek invests £154 500 for 2 years at 4% per year compound interest.

Work out the value of the investment at the end of 2 years.

(3 marks)

Q4**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 3 marks

7 Katy invests £200 000 in a savings account for 4 years.

The account pays compound interest at a rate of 1.5 % per annum.

Calculate the total amount of interest Katy will get at the end of 4 years.

(3 marks)

Q5**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 3 marks

12 Aayush invests 18 000 rupees for 3 years at a rate of 4% per year compound interest.

Work out the total amount of interest Aayush has received by the end of 3 years.

Give your answer correct to the nearest rupee.

(3 marks)

PAST PAPERS

Q6**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q8

CH 1 • TOPIC 7 3 marks

8 Harold bought an antique clock for £1200
The clock increased in value by 8% per year.

Find the value of the clock exactly 3 years after Harold bought the clock.

Give your answer correct to the nearest £.

£.....

(Total for Question 8 is 3 marks)

Q7**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q7

CH 1 • TOPIC 7 3 marks

- 7 Kuro invests 50 000 yen for 3 years in a savings account.
She gets 2.4% per year compound interest.

Work out how much money Kuro will have in her savings account at the end of the 3 years.
Give your answer correct to the nearest yen.

..... yen

(Total for Question 7 is 3 marks)

Q8 **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1H • Q7**CH 1 • TOPIC 7** **6 marks**

- 7 Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

- (a) Work out the value of Chen's investment at the end of 3 years.
Give your answer to the nearest yuan.

..... yuan

Wang invested P yuan.

The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

- (b) Work out the value of P .

$P =$

(Total for Question 7 is 6 marks)

Q9**PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 2H • Q11

CH 1 • TOPIC 7 6 marks

11 Chengbo sold a house for 180 000 yuan.

The amount for which he sold the house is 24% more than the amount he paid for the house.

- (a) Work out how much Chengbo paid for the house.
Give your answer correct to 3 significant figures.

..... yuan

Zhi bought a house on 1st January 2017

When she bought the house, its value was 120 000 yuan.

The value of the house increased by 1.8% per year.

- (b) Work out the value of Zhi's house on 1st January 2020
Give your answer correct to 3 significant figures.

..... yuan

(Total for Question 11 is 6 marks)

Q10**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2HR • Q5

CH 1 • TOPIC 7 3 marks

- 5 Shane invests 7200 dollars for 3 years in a savings account.
He gets 2.5% per year compound interest.

How much money will Shane have in his savings account at the end of 3 years?
Give your answer to the nearest dollar.

..... dollars

(Total for Question 5 is 3 marks)

Q11**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Pasha invests 50000 dollars in a savings account for 4 years.
He gets 1.3% per year compound interest.

Work out how much money Pasha will have in his savings account at the end of 4 years.
Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 9 is 3 marks)

Q12**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2H • Q8

CH 1 • TOPIC 7 3 marks

- 8 Charlotte buys a painting for \$680
The value of the painting increases by 4% each year.

Work out the value of the painting at the end of 3 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

Q13**PAST PAPER**

Edexcel 4MA1 • June 2024 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of
4 years.

Give your answer correct to the nearest koruna.

..... koruna

(Total for Question 9 is 3 marks)

Q14**PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q5

CH 1 • TOPIC 7 3 marks

- 5 Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account
at the end of 4 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 5 is 3 marks)

Q15**PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 2H • Q9

CH 1 • TOPIC 7 3 marks

- 9 Nisha invests 20 000 euros for 3 years in a savings account.
She gets 3.5% per year compound interest.

Work out how much money Nisha will have in her savings account
at the end of the 3 years.

Give your answer correct to the nearest euro.

..... euros

(Total for Question 9 is 3 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 2H • Q7

CH 1 • TOPIC 7 3 marks

- 7 Tim buys a bracelet for 4000 Swiss francs.
The value of the bracelet increases by 7% each year.

Work out the value of the bracelet at the end of 3 years.
Give your answer correct to the nearest Swiss franc.

..... Swiss francs

(Total for Question 7 is 3 marks)

02 Depreciation

Same formula, multiplier below 1. The value goes down by a percentage of the **current** value each year.

WHEN YOU SEE → YOU DO

“depreciates by 19% each year”

→ start $\times 0.81^n$

“the total percentage it had depreciated by”

→ $1 - 0.85^2 = 0.2775$: 27.75%

“lost more than half its value?”

→ compare the final value with half the start

different rates each year

→ multiply each year's multiplier

ZONE GROWTH

multiplier > 1

1.04, 1.025, 1.35

ZONE DEPRECIATION

multiplier < 1

0.81, 0.85, 0.8

WORKED MODEL • JANUARY 2020 P2HR

A car bought for \$20 000 depreciates by 19% each year. Value after 3 years?

1 multiplier $1 - 0.19 = 0.81$

down 19%

2 $20\,000 \times 0.81^3 = 10\,628.82$

power 3

ANSWER \$10 629

× **TRAP** $19\% \times 3 = 57\%$ off is wrong: after the first year you lose 19% of a smaller value.

IDEA 02

Practise: Depreciation

My questions first, then selected practice, then real past papers.

Q17 DR ESLAM Depreciation

CH 1 • TOPIC 7 3 marks

A machine worth £8000 depreciates by 10% each year. Find its value after 3 years.

SELECTED PRACTICE

Q18**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 4 marks

- 4 This notice was in a car magazine.

Most new cars lose more than half of their value in the first three years

Paul bought a new car.

The value of the car was £15 000

In the first year, the value of the car depreciated by 23%.

After the first year, the value of the car depreciated by 18% each year.

Work out if Paul's car lost more than half of its value by the end of three years.

(4 marks)

PAST PAPERS

Q19**PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q8

CH 1 • TOPIC 7 3 marks

- 8 Hamish buys a new car for \$20 000
The car depreciates in value by 19% each year.

Work out the value of the car at the end of 3 years.
Give your answer to the nearest \$.

\$.....

(Total for Question 8 is 3 marks)**Q20****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q8

CH 1 • TOPIC 7 3 marks

- 8 Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.

Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$.....

(Total for Question 8 is 3 marks)

Q21 PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q6

CH 1 • TOPIC 7 6 marks

6 Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	Service charge band
$x \geq 700\,000$	A
$600\,000 \leq x < 700\,000$	B
$500\,000 \leq x < 600\,000$	C
$400\,000 \leq x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

- (a) Has the annual service charge band changed for Asha's apartment?
Show your working clearly.

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

- (b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

..... %

(Total for Question 6 is 6 marks)

Q22**PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1HR • Q9

CH 1 • TOPIC 7 3 marks

- 9 Kazi buys a car for 700 000 taka.
The value of the car depreciates by 12% each year.

Work out the value of the car at the end of 3 years.
Give your answer correct to the nearest taka.

..... taka

(Total for Question 9 is 3 marks)

Q23**PAST PAPER**

Edexcel 4MA1 • November 2023 • Paper 2H • Q7

CH 1 • TOPIC 7 3 marks

- 7 Hermione buys a boat for \$26 800
The value of the boat depreciates by 8% each year.

Work out the value of the boat at the end of 3 years.
Give your answer correct to the nearest dollar.

\$.....

(Total for Question 7 is 3 marks)

Q24**PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q5

CH 1 • TOPIC 7 6 marks

5 Otis sells ice creams.

On Friday, Otis sells 75 ice creams.

On Saturday, Otis sells 87 ice creams.

- (a) Work out the percentage increase in the number of ice creams Otis sells from Friday to Saturday.

..... %

Claudia buys an ice cream machine for 960 Swiss francs.

The value of the ice cream machine depreciates by 20% each year.

- (b) Work out the value of the ice cream machine at the end of 3 years.

..... Swiss francs

(Total for Question 5 is 6 marks)

03 Compare and Changing Rates

Different rates in different years: one multiplier for each year, multiply them all. To compare offers, work out each final amount (or interest), then decide in a sentence.

WHEN YOU SEE → YOU DO

“4% for the first year, then 1.5% each year”

→ $\times 1.04 \times 1.015 \times 1.015 \dots$

“which bank should she choose?”

→ final amount of each, then choose

“receive 1150 dollars interest each year”

→ simple: 3×1150

“the average of 1% and 5% is 3% ...”

→ show $1.01 \times 1.05 \neq 1.03^2$

WORKED MODEL • NOVEMBER 2021 P2H

\$25 000: Cyclone Bank 4.5% compound for 3 years; Tornado Bank \$1150 interest each year. How much more interest from Cyclone?

1 $25\,000 \times 1.045^3 - 25\,000 = 3529.15$

compound interest

2 $3 \times 1150 = 3450$

simple interest

ANSWER \$79 more

IDEA 03

Practise: Compare and Changing Rates

My questions first, then selected practice, then real past papers.

Q25 **DR ESLAM** Changing rate

CH 1 • TOPIC 7 3 marks

£1000 earns 5% in year 1 and 2% in year 2 (compound). Find the total after 2 years.

SELECTED PRACTICE

Q26**PRACTICE**

Practice • Medium

CH 1 • TOPIC 7 4 marks

3 Peter has £20 000 to invest in a savings account for 2 years.

He finds information about two savings accounts.

Bonus Saver Compound interest 4% for the first year then 1.5% each year	Fixed Rate Compound interest 2.5% each year
--	--

Peter wants to have as much money as possible in his savings account at the end of 2 years.

Which of these savings accounts should he choose?

(4 marks)

Q27**PRACTICE**

Practice · Hard

CH 1 • TOPIC 7 4 marks

- 1 Viv wants to invest £2000 for 2 years in the same bank.

The International Bank

Compound Interest

4% for the first year
1% for each extra year

The Friendly Bank

Compound Interest

5% for the first year
0.5% for each extra year

At the end of 2 years, Viv wants to have as much money as possible.

Which bank should she invest her £2000 in?

(4 marks)

Q28**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 6 marks

- 3 (a)** Katie invests £200 in a savings account for 2 years.

The account pays compound interest at an annual rate of

3.3% for the first year 1.5% for the second year

Work out the total amount of money in Katie's account at the end of 2 years.

(3 marks)

- (b)** Katie travels to work by train. The cost of her weekly train ticket increases by 12.5% to £225

Katie's weekly pay increases by 5% to £535.50

Compare the increase in the amount of money Katie has to pay for her weekly train ticket with the increase in her weekly pay.

(3 marks)

Q29

PRACTICE

Practice • Hard

CH 1 • TOPIC 76 marks

PRACTISE • 1.7

10 (a) Here are the interest rates for two accounts.

Account A	Account B
Interest: 3% per year compound interest.	Interest: 4% for the first year, 3% for the second year and 2% for the third year.
No withdrawals until the end of three years.	Withdrawals allowed at any time.

Derrick has £10000 he wants to invest.
Calculate which account would give him most money if he invests his money for 3 years.
Give the difference in the interest to the nearest penny.

Account by p
(5 marks)

(b) Explain why he might **not** want to use Account A.
(1 mark)

PAST PAPERS

Q30**PAST PAPER**

Edexcel 4MA1 • November 2021 • Paper 2H • Q7

CH 1 • TOPIC 7**4 marks**

- 7 Ali and Badia each have 25 000 dollars to invest.

Cyclone Bank	Tornado Bank
Invest 25 000 dollars 4.5% compound interest per year for 3 years	Invest 25 000 dollars Receive 1150 dollars interest each year for 3 years

Ali invests in the Cyclone Bank for 3 years.

Badia invests in the Tornado Bank for 3 years.

By the end of the 3 years, Ali will have received more interest than Badia.

How much more?

Show your working clearly.

Give your answer correct to the nearest dollar.

..... dollars

(Total for Question 7 is 4 marks)

Q31**PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q11

CH 1 • TOPIC 7 5 marks

- 11 Zhou invests some money for 2 years.
He invests \$2500 with Bank A and \$3000 with Bank B.

Bank A		
Invests \$2500		
amount of money invested	:	total amount of interest after 2 years = 20 : 3
Bank B		
Invests \$3000		
4% per year compound interest for 2 years		

Zhou receives more **interest** from Bank A than from Bank B.

How much more?

\$

(Total for Question 11 is 5 marks)

04 Work Backwards: Rate, Start or Time

Three things in the formula; you are given two. Divide to find the start, take a root to find the rate, try years to find the time.

WHEN YOU SEE → YOU DO

“find the value of P ” (the start)

→ amount $\div$ multiplier ^{n}

“find x ” (the rate)

→ multiplier = $\sqrt[n]{\text{amount} \div \text{start}}$, then -1 , $\times 100$

“ $x\%$ in the second and third year”

→ divide out the known year first, then $\sqrt{\quad}$

“the least time in whole years”

→ try $n = 1, 2, 3, \dots$ until it passes

Amount $\div$ start

Root (years)

Minus 1

$\times 100$

WORKED MODEL • JANUARY 2022 P1H

200 000 yen, 1.8% in year 1 and $x\%$ in years 2 and 3. Total 209 754 yen. Find x .

$$1 \quad \frac{209\,754}{200\,000 \times 1.018} = 1.030225$$

divide out year 1

$$2 \quad \sqrt{1.030225} = 1.015$$

two years → square root

ANSWER $x = 1.5$

× **TRAP** Dividing the percentage by the years ($6.1\% \div 3$). Rates multiply — you need the root.

IDEA 04

Practise: Work Backwards: Rate, Start or Time

My questions first, then selected practice, then real past papers.

Q32

DR ESLAM

Find the rate

CH 1 • TOPIC 7

3 marks

An investment grows from £4000 to £4840 in 2 years at $r\%$ compound interest. Find r .

SELECTED PRACTICE

Q33**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 6 marks

- 2 (a)** The population of a city increased by 5.2% for the year 2014

At the beginning of 2015 the population of the city was 1 560 000

Lin assumes that the population will continue to increase at a constant rate of 5.2% each year.

Use Lin's assumption to estimate the population of the city at the beginning of 2017
Give your answer correct to 3 significant figures.

(3 marks)

- (b)** i) Use Lin's assumption to work out the year in which the population of the city will reach 2 000 000
ii) If Lin's assumption about the rate of increase of the population is too low, how might this affect your answer to (b)(i)?

(3 marks)

Q34**PRACTICE**

Practice • Hard

CH 1 • TOPIC 7 **3 marks**

9 Mirek invests £6000 at a compound interest rate of 1.5% per year.

He wants to earn more than £1000 interest.

Work out the **least** time, in whole years, that this will take.

.....years

(3 marks)

Q35**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 7 8 marks

4 (a) Katy invests £2000 in a savings account for 3 years.

The account pays compound interest at an annual rate of

2.5% for the first year

$x\%$ for the second year

$x\%$ for the third year

There is a total amount of £2124.46 in the savings account at the end of 3 years.

Work out the rate of interest in the second year.

(4 marks)

(b) Katy goes to work by train.

The cost of her weekly train ticket increases by 12.5% to £225

Work out the cost of her weekly train ticket before this increase.

(4 marks)

Q36**PRACTICE**

Practice • Very Hard

CH 1 • TOPIC 7 3 marks

- 8 Hans invests \$550 at a rate of $x\%$ per year compound interest.
At the end of 10 years, the value of the investment is \$638.30, correct to the nearest cent.

Find the value of x .

$x = \dots\dots\dots$ [3]

(3 marks)

PAST PAPERS**Q37****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 1H • Q11

CH 1 • TOPIC 7 3 marks

- 11 Max invests \$6000 in a savings account for 3 years.
The account pays compound interest at a rate of 1.5% per year for the first 2 years.

The compound interest rate changes for the third year.
At the end of 3 years, there is a total of \$6311.16 in the account.

Work out the compound interest rate for the third year.
Give your answer correct to 1 decimal place.

$\dots\dots\dots\%$

(Total for Question 11 is 3 marks)

Q38**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q13

CH 1 • TOPIC 7 3 marks

- 13 Jan invests \$8000 in a savings account.
The account pays compound interest at a rate of $x\%$ per year.
At the end of 6 years, there is a total of \$8877.62 in the account.
Work out the value of x .
Give your answer correct to 2 decimal places.

 $x = \dots\dots\dots$ **(Total for Question 13 is 3 marks)**

Q39**PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 2HR • Q14

CH 1 • TOPIC 7 4 marks

14 Simon bought a house at the beginning of 2018

The value of Simon's house had decreased by 15% by the end of 2018

The house increased in value during both 2019 and 2020

The percentage increases in the value of the house during 2019 and 2020 were the same.

The value of Simon's house at the end of 2020 was 2.85% greater than the amount he paid for his house at the beginning of 2018

Calculate the percentage increase in the value of the house during 2019

.....%

(Total for Question 14 is 4 marks)

Q40**PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1H • Q11

CH 1 • TOPIC 7 3 marks

- 11 Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and $x\%$ for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of x

Give your answer correct to one decimal place.

 $x = \dots\dots\dots$

(Total for Question 11 is 3 marks)

Q41**PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q13

CH 1 • TOPIC 7 3 marks

- 13 Feruzi invests 80 000 Kenyan shillings (KES)

He invests the money for 3 years at $x\%$ compound interest each year.

At the end of 3 years, the total interest he receives is 6151.25 KES

Work out the value of x

 $x = \dots\dots\dots$

(Total for Question 13 is 3 marks)

Q42 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 1H • Q13

CH 1 • TOPIC 7 3 marks

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x

$x =$

(Total for Question 13 is 3 marks)

Q43 **PAST PAPER** Edexcel 4MA1 • June 2025 • Paper 2HR • Q12

CH 1 • TOPIC 7 3 marks

12 Osman buys a car for \$16 000

The car depreciates at a rate of 12% each year for the first 2 years.

In the third year, the car depreciates at a rate of $x\%$

At the end of 3 years, the value of the car is \$11 461.12

Work out the value of x

.....

(Total for Question 12 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

ZONE GROWTH

$$\times(1 + r)^n$$

ZONE DEPRECIATION

$$\times(1 - r)^n$$

- | | | |
|-----------|-------------------------|---|
| 01 | Growth | Start $\times$ multiplier ⁿ ; interest = amount – start. |
| 02 | Depreciation | Multiplier below 1; the % is never multiplied by the years. |
| 03 | Compare / change | One multiplier per year; decide with the final amounts. |
| 04 | Backwards | Start: divide. Rate: root. Time: try $n = 1, 2, 3 \dots$ |

BOTTOM LINE amount = start $\times$ multiplierⁿ $\rightarrow$ fill two boxes, find the third.

Answers

Compound Interest and Depreciation · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • COMPOUND GROWTH

Q1	Dr Eslam • Growth	$5000 \times 1.03^2 = £5304.50$
Q2	Practice • Medium	£8112
Q3	Practice • Medium	(a) 3% (b) £167 107.20
Q4	Practice • Medium	£12 272.71
Q5	Practice • Medium	2248 rupees
Q6	Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q8	£1512
Q7	Edexcel 4MA1 • January 2021 • Paper 1HR • Q7	53 687 yen
Q8	Edexcel 4MA1 • January 2021 • Paper 1H • Q7	(a) 44 349 yuan (b) $P = 32\,600$
Q9	Edexcel 4MA1 • May 2021 • Paper 2H • Q11	(a) 145 000 yuan (b) 127 000 yuan
Q10	Edexcel 4MA1 • January 2022 • Paper 2HR • Q5	\$7754
Q11	Edexcel 4MA1 • June 2022 • Paper 2H • Q9	\$52 651
Q12	Edexcel 4MA1 • June 2023 • Paper 2H • Q8	\$765
Q13	Edexcel 4MA1 • June 2024 • Paper 2H • Q9	4948 koruna
Q14	Edexcel 4MA1 • May 2024 • Paper 1HR • Q5	5740 euros
Q15	Edexcel 4MA1 • November 2024 • Paper 2H • Q9	22 174 euros
Q16	Edexcel 4MA1 • June 2025 • Paper 2H • Q7	4900 Swiss francs

IDEA 02 • DEPRECIATION

- Q17** Dr Eslam • Depreciation $8000 \times 0.9^3 = £5832$
- Q18** Practice • Medium $15\,000 \times 0.77 \times 0.82^2 = £7766 > £7500$: it lost less than half
- Q19** Edexcel 4MA1 • January 2020 • Paper 2HR • Q8 \$10 629
- Q20** Edexcel 4MA1 • June 2022 • Paper 1HR • Q8 \$9396
- Q21** Edexcel 4MA1 • June 2022 • Paper 1H • Q6 (a) 2020 value 610 000: band B both years — No (b) 27.75%
- Q22** Edexcel 4MA1 • May 2023 • Paper 1HR • Q9 477 030 taka
- Q23** Edexcel 4MA1 • November 2023 • Paper 2H • Q7 \$20 869
- Q24** Edexcel 4MA1 • November 2025 • Paper 2H • Q5 (a) 16% (b) 491.52 Swiss francs

IDEA 03 • COMPARE AND CHANGING RATES

- Q25** Dr Eslam • Changing rate $1000 \times 1.05 \times 1.02 = \text{£}1071$
- Q26** Practice • Medium Bonus £21 112, Fixed £21 012.50: Bonus Saver
- Q27** Practice • Hard International £2100.80, Friendly £2110.50: The Friendly Bank
- Q28** Practice • Hard (a) £209.70 (b) ticket up £25.00, pay up £25.50
- Q29** Practice • Hard (a) A by 103p (£10 927.27 vs £10 926.24) (b) he cannot withdraw for 3 years
- Q30** Edexcel 4MA1 • November 2021 • Paper 2H • Q7 3529.15 – 3450: \$79
- Q31** Edexcel 4MA1 • June 2025 • Paper 1HR • Q11 A: \$375, B: \$244.80: \$130.20 more

IDEA 04 • WORK BACKWARDS: RATE, START OR TIME

- Q32** Dr Eslam • Find the rate $\sqrt{1.21} = 1.1: r = 10$
- Q33** Practice • Hard (a) 1730 000 (b)(i) 2020 (ii) it would reach 2 000 000 sooner
- Q34** Practice • Hard 11 years
- Q35** Practice • Very Hard (a) 1.8% (b) £200
- Q36** Practice • Very Hard $x = 1.5$
- Q37** Edexcel 4MA1 • January 2020 • Paper 1H • Q11 2.1%
- Q38** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q13 $x = 1.75$
- Q39** Edexcel 4MA1 • January 2021 • Paper 2HR • Q14 10%
- Q40** Edexcel 4MA1 • January 2022 • Paper 1H • Q11 $x = 1.5$
- Q41** Edexcel 4MA1 • June 2023 • Paper 2HR • Q13 $x = 2.5$
- Q42** Edexcel 4MA1 • June 2025 • Paper 1H • Q13 $x = 3.75$
- Q43** Edexcel 4MA1 • June 2025 • Paper 2HR • Q12 $x = 7.5$